CIA-1

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Fundamentals of machine learning.

CO1/25 | CO2/18

43/50

1. Soln:

Given:

Prevalence: $P(C) = 0.03$ $P(\bar{C}) = 0.97$.

Sensitivity: $P(+/C) = 0.90$ $P(+/\bar{C}) = 0.94$.

False-Positive Probability:

$$P(+/\bar{C}) = 1 - 0.94 = 0.06$$

False-negative Probability

$$P(-/C) = 1 - 0.90 = 0.10$$

Costs:

- with holding treatment from diseased patient = 50
- Treating healthy patient = 2
- correct decision = 0.

1. Probability of positive result:

$$P(+) = P(+/C)\,P(C) + P(+/\bar{C})\,P(\bar{C})$$
$$= 0.90 \times 0.03 + 0.06\,(0.97)$$
$$= 0.0852.$$

2. Posterior Probability $P(C/+)$

$$P(C/+) = \frac{P(+/C)\,P(C)}{P(+)} = \frac{0.90 \times 0.03}{P(+)} = 0.3169.$$

3.

3. $P(+|-)$

$P(-) = 1 - 0.0852 = 0.9148.$

$\therefore P(c|-) = \frac{P(-|c)P(c)}{P(-)} = \frac{0.10 \times 0.03}{0.9148}$

4. Conditional risk for positive result:

Action 1: Treat $R(T|+) = 2P(\bar{c}|+) = 2(1 - 0.3169)$

$= 1.3662.$

Action 2: Donot treat $R(NT|+) = 50 P(c|+) = 50(0.3169)$

$= 15.845$

Minimum risk: Treat the patient.

5. Conditional risk for negative result.

Action 1: Treat

$R(T|-) = 2(1 - 0.00328) = 1.9934.$

Action 2: Donot treat $R(NT|-) = 50(0.00328) = 0.164.$

Minimum risk: Donot treat

6. $R = P(+)R(T|+) + P(-)R(NT|-)$

$= 0.0852(1.3662) + 0.9148(0.164)$

$= 0.2664.$

7. $P(c|+,+) = \frac{0.3169(0.90)}{0.3169(0.90) + 0.6831(0.06)} = 0.8749$

$=$

8. Minimum error (0-1) loss rule: decide c only if $P(c|*) > 0.5$

Since $P(c|+) = 0.3169 < 0.5$, the rule never treat; its expected cost under the real loss matrix $= 50 \times 0.03 = 1.5$ per patient

The minimum risk rule reduces expected cost from 1.5 to 0.2664, a reduction 82.24%.

Interpretation and Conclusion:

Minimum risk decision theory is more suitable for hospitals because missing a diseased patient is much more costly than unnecessarily treating the healthy patient.

2. Soh

1. Marginal distribution of x and Y.

$$P(x) = \sum_y P(x,y) \qquad P(y) = \sum_x P(x,y)$$

$P(\text{low}) = 0.30 + 0.02 = 0.32$

$P(\text{medium}) = 0.28 + 0.10 = 0.38$

$P(\text{High}) = 0.10 + 0.20 = 0.30$

$\therefore P(x) = (0.32, 0.38, 0.30)$

$P(\text{No}) = 0.30 + 0.28 + 0.10 = 0.68$

$P(\text{Yes}) = 0.02 + 0.10 + 0.20 = 0.32.$

2. $H(x) = -\sum P(x) \log_2 P(x) = -[(0.32 \log_2 0.32) + 0.38 \log_2(0.38) + 0.30 \log_2(0.3)$

$= 1.5776$ bits

$H(Y) = -[0.68 \log_2 0.68) + 0.32 \log_2(0.32)]$

$= 0.9044$ bits.

3. Joint entropy: $H(x,Y) = -\sum P(x,y) \log_2 P(x,y)$

$= -[0.30 \log_2(0.30) + 0.02 \log_2(0.02)$

$= 2.2770$ bits

3. Conditional entropy: $H(Y|x) = H(x,Y) - H(x) = 2.2770 - 1.5776$

$= 0.6994$ bits.

4. Mutual information: $I(x;Y) = H(x) + H(Y) - H(x,Y)$

$= 1.5776 + 0.9044 - 2.2770$

$= 0.2050$ bits.

5. Cross entropy: $Q = (0.5, 0.3, 0.2)$, $P(x,Q) = -\sum P(x) \log_2 Q(x)$

$= -[0.32 \log_2(0.5) +$

$= +0.38 \log_2(0.3) + 0.38 \log_2(0.2)$

$= 1.6766$ bits.

6. KL divergence:

$$D_{KL}(P||Q) = \sum P(x) \log_2 P(x)/Q(x)$$

$$= 0.32 \log_2 0.32/0.5 + 0.38 \log_2 0.38/0.3 + 0.30 \log_2 0.30/0.2$$

$$= 0.09905 \text{ bits}$$

7. $P(\text{Yes} | \text{High}) = \frac{P(\text{High}, \text{Yes})}{P(\text{High})} = \frac{0.20}{0.30} = 0.6667.$

$I(X; Y) = 0.2050.$

Therefore, network load alone is not a sufficiently strong predictor of link faults

3. Soln

Given:

Samples $x = \{48, 52, 55, 47, 53, 51\}$ So, $n = 6$.

1) ML Mean $\mu = \frac{\sum x}{n} = \frac{48+52+55+47+53+51}{6} = 51.0^{\circ}C$

The ML estimate of the Gaussian mean is the sample mean.

2. Squared deviation:

$$= (-3)^2 + 1^2 + 4^2 + (-4)^2 + 2^2 + 0^2 = 9+1+16+16+4$$

$$= 46.$$

ML variance $\sigma^2 = 46/6 = 7.667$. $\sigma = 2.769^{\circ}C$.

3. Unbiased variance: $S^2 = \frac{\sum (x_i - \bar{x})^2}{(n-1)} = 46/5 = 9.20$

$S = 3.033^{\circ}C$

Dividing by n-1 corrects the bias, because $E[\sigma^2_{ML}] = (n-1)/n\,\sigma^2 = 5/6\,\sigma^2$, ie. the ML estimate is 16.7% too small on average for n=6.

4. $P(X \geq 55)$; $z = (55-51)/2.769 = 1.445$;

$P = 1 - \Phi(1.445) = 0.0743$.

Table values $\Phi(1.44) = 0.925$ and $\Phi(1.45) = 0.9265$ given 0.074 - 0.075. (accept 0.074 ± 0.001). With $s = 3.033$ the tail become 0.0936. Showing the effect of the bias.

5. $P(48 < X < 54)$; $z = \pm(3/2.769) = \pm 1.083$, $P = 2\Phi(1.083) - 1$

$= 0.7214$

Accept 0.72 ± 0.01 using $\Phi(1.08) = 0.8599$.

Conclusion:

- The ML estimates are $\mu = 51.0$ and $\sigma^2 = 7.667\ c^2$. The unbiased variance is larger (9.20). For small samples the ML variance underestimate the true variance and ∴ underestimates tail probabilities. $P(0.074$ against 0.0936 for $P(X \geq 55)$.

ii)

1. Mean $\bar{x} = 30$; $\bar{y} = (25+38+52+61+79)/5 = 51$.

$S_{xx} = \Sigma(x-\bar{x})^2 = 400+100+0+100+400 = 1,000$

$S_{xy} = \Sigma(x-\bar{x})(y-\bar{y}) = 520+130+0+100+500 = 1310$

The normal equations reduce to $w_1 = S_{xy}/S_{xx}$ and $w_0 = \bar{y} - w_1\bar{x}$

2. $w_1 = 1310/1000 = 1.31$ mks per %. $w_0 = 51 - 1.31 \times 30 = 11.7$ mm

Fitted model: $\hat{y} = 11.7 + 1.31x$.

3 Fitted values = 24.8, 37.9, 51.0, 64.1, 77.2; residuals: 0.2, 0.1, 1.0, -3.1, 1.8; $SSE = 0.04 + 0.01 + 1.00 + 9.61 + 3.24$

$= 13.90$.

SSE measures the unexplained variation.

4. SST = 676 + 169 + 1 + 100 + 784 $R^2 = 1 - 13.90/1730$
= 1730 $= 0.9920$

The line explains 99.2% of the variation in response time.

5. Prediction at $x = 60$: $\hat{y} = 11.7 + 1.31 \times 60$
$= 90.3$ ms,

$x = 60$ lies outside the training range 10–50, so this is extrapolation

6. Ridge : $w_1 = S_{xy} / (S_{xx} + \lambda) = 1310 / 1600$ $w_0 = 51 - 0.8733 \times 30$
$= 0.8733$ $= 24.8$.

$\hat{y}(60) = 24.8 + 52.4 = 77.2$ ms.

SSE = 204.8 ; $R^2 = 0.8817$. So L_2 shrinks the slope by 33%.
and SSE increases about 14.7 times.

7. Conclusion :

Ordinary linear regression predicts 90.3ms, whereas ridge regression predicts 77.2ms. Ridge regularisation reduce the slope & may reduce sensitivity overfitting. However, with this small dataset, neither model can be declared more reliable without additional validation data.

4. Soln:

1. Priors : $P(spam) = 40/100 = 0.40$ $P(legit) = 60/100 = 0.60$
class frequencies in the training set.

2. Smoothed likelihoods, d_{spam} (denominator 42) : offer $29/42$
$= 0.6905$

free $25/42 = 0.5952$

meeting $34/62 = 0.5484$.

Adding 1 to the numerator and 2 to the denominator.

3. Smoothed likelihood : legitimate (denominator 62); offer 7/62 = 0.1129; free 4/62 = 0.0645; meeting 34/62 = 0.5484. The word "meeting" is typical of legitimate mail.

4. Scores for x = (offer = 1, free = 1, meeting = 0)

Spam = 0.40 × 0.6905 × 0.5952 × (1 − 0.1190) = 0.1448

Legit = 0.60 × 0.1129 × 0.0645 × (1 − 0.5484) = 0.00197

5. P(spam | x) = 0.1448 / 2 = 0.9866

P(legit | x) = 0.0134.

6. P(spam | x) 5/6 = 0833 since since 9.861 > 0.833 → Spam

7. Accuracy = 87.5%. Precision = 81.2%. Recall 75.1%.

F1-Score = 78.26%.

8. Total = 5 × FP + 1 × FN = 5 × 1 + 5 = 65 false = 7.14%.

Conclusion: Raise the threshold to cut fals positive, add more features or use a discriminative model.

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5. 1. Hiden : $z_1 = 0.100$

$z_2 = 0.550$.

2. Activities : $h_1 = \sigma(0.100) = 0.5250$

$h_2 = \sigma(0.550) = 0.6341$.

3. Output : $z_0 = 2.613$; $y = \sigma(2.613) = 0.5652$.

Output of the network before training

4. Error : $E = 1/2 (0.4350)^2 = 0.0946$

The network Under-Predicts the target